

While Loop, For Loop, and Function

Task 1: While Loop & Control Statements -Number Guessing Game

1. Set Up the Game:

```
import random

secret_number = random.randint(1, 10)

print(secret_number)
```

3

2. Prompt the User:

```
attempts = 3

guess = int(input("Guess the number (between 1 and 10): "))

print("Your guess is:", guess)
print("Attempts available:", attempts)
```

Guess the number (between 1 and 10): 5
Your guess is: 5
Attempts available: 3

3. Implement the Guessing Logic:

```
while attempts > 0:
    guess = int(input("Guess the number (between 1 and 10): "))
    attempts = attempts - 1
    print("Attempts remaining:", attempts)
```

Guess the number (between 1 and 10): 5
Attempts remaining: 2
Guess the number (between 1 and 10): 7
Attempts remaining: 1
Guess the number (between 1 and 10): 1
Attempts remaining: 0

```

while attempts > 0:
    guess = int(input("Guess the number (between 1 and 10): "))

    if guess < 1 or guess > 10:
        print("Your guess is out of range. Please guess a number between 1 and 10.")
        continue

    break

```

```

Guess the number (between 1 and 10): 15
Your guess is out of range. Please guess a number between 1 and 10.
Guess the number (between 1 and 10): 7

```

4. Control Statements:

```

attempts = 3

while attempts > 0:
    guess = int(input("Guess the number (between 1 and 10): "))

    if guess < 1 or guess > 10:
        print("Your guess is out of range. Please guess a number between 1 and 10.")
        continue

    attempts = attempts - 1

    if guess == secret_number:
        print("Congratulations! You guessed the correct number.")
        break
    elif guess > secret_number:
        print("Too high. Try again.")
    else:
        print("Too low. Try again.")

```

```

Guess the number (between 1 and 10): 7
Too high. Try again.
Guess the number (between 1 and 10): 5
Too high. Try again.
Guess the number (between 1 and 10): 3
Congratulations! You guessed the correct number.

```

Task 2:- For Loop - Multiplication Table Generator

```
number = int(input("Enter the number for which you want the multiplication table: "))
```

Enter the number for which you want the multiplication table: 5

```
for i in range(1, 11):  
    print(i)
```

1
2
3
4
5
6
7
8
9
10

```
for i in range(1, 11):  
    result = number * i  
    print(result)
```

5
10
15
20
25
30
35
40
45
50

```
for i in range(1, 11):  
    result = number * i  
    print(number, "x", i, "=", result)
```

```
5 x 1 = 5  
5 x 2 = 10  
5 x 3 = 15  
5 x 4 = 20  
5 x 5 = 25  
5 x 6 = 30  
5 x 7 = 35  
5 x 8 = 40  
5 x 9 = 45  
5 x 10 = 50
```

```
number = int(input("Enter the number for which you want the multiplication table: "))  
  
for i in range(1, 11):  
    result = number * i  
    print(number, "x", i, "=", result)
```

```
Enter the number for which you want the multiplication table: 5  
5 x 1 = 5  
5 x 2 = 10  
5 x 3 = 15  
5 x 4 = 20  
5 x 5 = 25  
5 x 6 = 30  
5 x 7 = 35  
5 x 8 = 40  
5 x 9 = 45  
5 x 10 = 50
```

Task 3: Function- BMI Calculator

```
def calculate_bmi(weight, height):  
    bmi = weight / (height ** 2)  
    return bmi
```

```
weight = float(input("Enter your weight in kg: "))
```

```
Enter your weight in kg: 70
```

```
height = float(input("Enter your height in meters: "))
```

Enter your height in meters: 2.5

```
print(f"Your BMI is: {bmi:.2f}")
```

Your BMI is: 11.20